KILNDECK · PRODUCT REQUIREMENTS

# KilnDeck product requirements and specifications

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## DOCUMENT CONTROL

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| Document ID | KND-PRS-001 | Version | 0.4 |
| Status | Draft | Date | 2026-10-30 |
| Author | Amish Chadha | License | CERN-OHL-S-2.0 |
| Project TRL | TRL 3 Proof of concept | | |
| Repository | github.com/BoujeeEnjinia1701/kilndeck | | |
| Source | docs/03-requirements.md @ 9f92bd8 | | |

## REVISION HISTORY

| Version | Date | Author | Change |
|---|---|---|---|
| 0.4 | 2026-10-30 | Amish Chadha | PR-012 limit changed from 40 kg to 33 kg and PR-016 module length from 3 m to 2.4 m (KND-DEC-001 D14, decided 2026-10-30); PR-012 not met by the end truss modules; working limit of 7 m/s with clamps at every shift end (D11); statuses and cost from KND-CAL-001 v0.3 |
| 0.3 | 2026-10-28 | Amish Chadha | Kit 1.8 migration to KND-PRS-001; R1 to R13 become PR-001 to PR-013 with tolerance, units and the user requirement each traces to; PR-014 (storm hold, KND-DEC-001 D12) and PR-015 to PR-017 (build, relaying and upkeep) added; R11 cost now USD 2,787 with the storm clamps |
| 0.2 | 2026-10-03 | Amish Chadha | R2 and R4 restated, R12 and R13 added (KND-DDR-001); status on paper from KND-CAL-001 for the constructable design |
| 0.1 | 2026-09-30 | Amish Chadha | Initial scaffold (as KND-REQ-001) |

# KilnDeck product requirements and specifications

On paper, the constructable design (option A, the rolling bridge deck) meets 12 of the 17 product requirements. PR-007 (floor temperature) is not met in full summer sun, PR-012 (no piece over 33 kg, since D14 of 2026-10-30) is not met by the two end truss modules (34.4 and 34.8 kg), PR-001 is met on the straight sides of the kiln but not on the curved ends, PR-008 is met only against an interim bearing limit until the partner kiln is surveyed, and PR-005 (dose repeatability) can only be judged by weighing doses at TRL 4. Every "met on paper" is a calculation in KND-CAL-001, not a test. Targets are proposals for review with a co-design partner. This document replaces KND-REQ-001; the old R numbers are in the "Was" column. Method: STANDARDS section 22.

*Table 1. Product requirements and specifications.*

| ID | Traces to | Category | Title | Requirement | Target | Tolerance | Units | Verification | Status | Was |
|---|---|---|---|---|---|---|---|---|---|---|
| PR-001 | UR-001, UR-003 | Usability | No steps on the crust | The fireman shall feed every hole in a round without stepping onto the crust | 0 | max | steps per round | Observed trial over 20 rounds, TRL 4 | Met on the straight sides; not met on the curved ends, which this prototype does not serve | R1 |
| PR-002 | UR-002 | Safety | Guardrail | The guardrail shall have a top rail 1.1 m above the floor, a knee rail at 0.55 m and a 150 mm toe board, hold 1 kN at the top rail and take 1.5 kN without yield | 1.1; 1.0; 1.5 | min | m; kN; kN | Analysis (KND-CAL-001 s7), TRL 3; proof load (CalRig), TRL 4 | Met on paper: 150 MPa at the proof load (yield 275 MPa) | R2 |
| PR-003 | UR-002 | Safety | Rated load | The deck shall carry 250 kg and a proof load of 1.5 times that | 250 | min | kg | Analysis (KND-CAL-001 s6), TRL 3; proof load, TRL 4 | Met on paper: 21 MPa in the chords at the proof load, with the carriage at mid-span | R3 |
| PR-004 | UR-004 | Usability | Hand wheel force | Moving the loaded deck on level track shall take no more than 200 N at the hand wheel rim | 200 | max | N | Analysis (KND-CAL-001 s10), TRL 3; force gauge, TRL 4 | Met on paper: 68 N level, 135 N on a 2 % grade | R4 |
| PR-005 | UR-006 | Performance | Fuel dose | Each rotor pocket shall deliver the set dose of dry coal within 15 % | 15 | max | % | Weigh 20 doses, TRL 4 | To verify at TRL 4; nominal 0.49 kg of coal per pocket | R5 |
| PR-006 | UR-003 | Performance | Hopper capacity | The hopper shall hold at | 1.0 | min | rounds | Analysis (KND- | Met on paper: 52 L, | R6 |

| ID | Traces to | Category | Title | Requirement | Target | Tolerance | Units | Verification | Status | Was |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | least one feeding round for one fireman | | | | CAL-001 s3), TRL 3 | about 42 kg of coal against about 30 kg a round (1.4 rounds) | |
| PR-007 | UR-005 | Safety | Floor temperature | The standing area shall be at 50 °C or less after 6 h on an operating kiln | 50 | max | °C | Analysis (KND-CAL-001 s11), TRL 3; contact thermometer log, TRL 4 | Not met in full sun: about 74 °C at 45 °C air; met at night (34 °C) and in shade (48 °C); the bare crust is about 150 °C | R7 |
| PR-008 | UR-007 | Physical architecture | Load on the kiln | The pressure under each track pad shall stay below the limit set by a structural survey of the partner kiln | 0.20 | max | MPa (interim) | Analysis (KND-CAL-001 s9), TRL 3; survey and load cell, TRL 4 | Met against the interim 0.20 MPa (0.154 MPa; 0.209 MPa at the 250 kg rated load); cannot be confirmed until the survey | R8 |
| PR-009 | UR-003 | Performance | Spout alignment | The spout outlet shall stop within 25 mm of the feed hole centre | 25 | max | mm | Analysis (KND-CAL-001 s12), TRL 3; measure at 10 holes, TRL 4 | Met on paper: about 17 mm along the kiln, 10 mm across | R9 |
| PR-010 | UR-013 | Safety | Parking brake | The lock pin shall hold the loaded deck on a 5 % grade with no creep | 5 | min | % grade | Analysis (KND-CAL-001 s10), TRL 3; inclined track test, TRL 4 | Met on paper: holds 767 N against 512 N | R10 |
| PR-011 | UR-010 | Cost | Cost | Value-engineering target for one deck and a demonstration length of track | 3,500 | target | USD | Costed BOM, TRL 3 | Estimated cost of the constructable design: USD 2,905 (USD | R11 |

| ID | Traces to | Category | Title | Requirement | Target | Tolerance | Units | Verification | Status | Was |
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| | | | | | | | | | 595 under the target) | |
| PR-012 | UR-008 | Packaging and transport | Piece mass | No piece carried up the kiln shall weigh more than 33 kg, the HSE two-person team-handling guide (KND-DEC-001 D14, decided 2026-10-30) | 33 | max | kg | Analysis from the model solids (KND-CAL-001 s5), TRL 3; weigh, TRL 4 | **Not met**: the end truss modules weigh 34.4 and 34.8 kg (1.4 and 1.8 kg over); every other piece is inside 33 kg (middle truss modules 32.9 kg, outer end truck 29.8 kg, track modules 26.7 kg, floor panels 24.8 kg); ways to close it in KND-DEC-001 D16 | R12 |
| PR-013 | UR-013 | Safety | Track retention | The deck shall not run off a track end or drop off a track | 0 | max | derailments | Analysis (KND-CAL-001 s9, s17), TRL 3; push and lift tests, TRL 4 | Met by design: end stops, buffers and derailment guards; the stops take 4.2 kN at a running speed of 0.28 m/s | R13 |
| PR-014 | UR-012, UR-013 | Safety | Storm hold | Parked and clamped, the deck shall not move along the kiln in a 28 m/s gust | 28 | min | m/s | Analysis (KND-CAL-001 s15), TRL 3; pull test on the clamps and anchors, TRL 4 | Met on paper with the storm clamps and tie-downs (KND-DEC-001 D12); anchor pull-out to assess at the partner kiln | |
| PR-015 | UR-011 | Materials | Local build | | 0 | max | | | | |

| ID | Traces to | Category | Title | Requirement | Target | Tolerance | Units | Verification | Status | Was |
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| | | | | Every made part shall be cut, drilled and stick welded from stock steel sections, plate and sheet | | | special processes | Inspection of the model and build plan (KND-BLD-001), TRL 3 | Met by design | |
| PR-016 | UR-009 | Packaging and transport | Track relaying | The track shall be laid ahead of the fire and lifted behind it in 2.4 m bolted modules | 2,400 | max | mm per module | Inspection (KND-BLD-001), TRL 3; relay trial, TRL 4 | Met by design | |
| PR-017 | UR-014 | Usability | Upkeep | Every chain shall be guarded and every bearing a catalogue part that a local fitter can replace | 0 | max | special parts | Inspection of the BOM and model, TRL 3 | Met by design | |

Categories: physical architecture, performance, materials, usability, packaging and transport, safety. Status: met on paper, met by design, at risk, not met, over target, withdrawn, to verify at TRL 4.

## Assumptions

- Kiln walls can carry the track pads without damage; to be confirmed kiln by kiln by a structural survey.
- The trench is about 6.0 m wide, the wall tops are at least 0.6 m wide and level with the crust within 100 mm, and feed holes are on a grid of about 1.0 m (KND-CAL-001, section 1).
- Firemen and owners accept a deck if it does not slow feeding.
- The track is laid in sections on the straight sides and moved with the fire.
- The deck is worked in winds up to 7 m/s on level track; above that, and at the end of every shift whatever the weather, it is clamped and tied down (KND-DEC-001 D11, decided 2026-10-30; KND-CAL-001, section 15).